Sexual Reproduction

Exam code: 0610 & 0970

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Gametes & Zygotes

Sexual Reproduction

- Sexual reproduction is **a process involving the fusion of the nuclei of two gametes (sex cells) to form a zygote (fertilised egg cell) and the production of offspring that are genetically different from each other**
- Fertilisation is defined as the **fusion of gamete nuclei**, and as each gamete comes from a different parent, there is variation in the offspring

Gametes

- A **gamete** is a sex cell (in animals: sperm and ovum; in plants pollen nucleus and ovum)
- Gametes differ from normal cells as they contain **half the number of chromosomes** found in other body cells - we say they have a **haploid nucleus**
- This is because they only contain **one copy of each chromosome**, rather than the two copies found in other body cells
- In human beings, a normal body cell contains **46 chromosomes** but each gamete contains **23 chromosomes**
- When the male and female gametes fuse, they become a **zygote** (fertilised egg cell)
- This contains the full **46 chromosomes**, half of which came from the father and half from the mother - we say the

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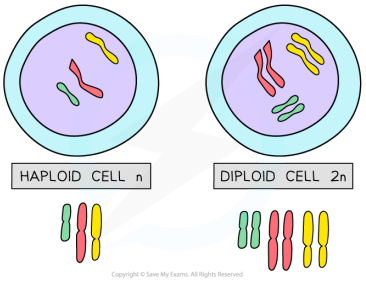
Haploid & Diploid Cells: Extended

- The nuclei of gametes are **haploid**
  - They contain half the number of chromosomes of a normal body cell
  - In humans, this is **23** chromosomes
- The nucleus of a zygote is **diploid**
  - It contains the same number of chromosomes as a normal body cell
  - In humans, this is **23 pairs** of chromosomes
  - The zygote continues to stay diploid as it grows into a fetus and embryo during pregnancy



Advantages & Disadvantages of Sexual Reproduction: Extended

ADVANTAGES	DISADVANTAGES
INCREASES GENETIC VARIATION	TAKES TIME AND ENERGY TO FIND MATES
THE SPECIES CAN ADAPT TO NEW ENVIRONMENTS DUE TO VARIATION, GIVING THEM A SURVIVAL ADVANTAGE	DIFFICULT FOR ISOLATED MEMBERS OF THE SPECIES TO REPRODUCE
DISEASE IS LESS LIKELY TO AFFECT POPULATION (DUE TO VARIATION)	

- Most crop plants reproduce sexually and this is an advantage as it means **variation is increased** and a genetic variant may be produced which is better able to cope with weather changes, or produces significantly higher yield
- The disadvantage is that the variation may lead to offspring that are **less successful** than the parent plant at growing well or producing a good harvest

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